

Veteran Mortality Patterns and Premature Mortality:

A Comparative Analysis of Recent Veterans Health Administration Users and Other Veterans (2018–2023)



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Why is this important?



Veteran mortality remains an important public health concern.



Although chronic diseases remain the leading causes of death, preventable causes such as suicide continue to contribute substantially to premature mortality.



Understanding mortality patterns can support:



evidence-based
healthcare
planning



suicide
prevention



population
health
management



resource
allocation



How do mortality patterns and Years of Potential Life Lost (YPLL) differ between Recent Veterans Health Administration (VHA) users and Other Veterans between 2018 and 2023?

Objectives

- ✓ Compare leading causes of death
- ✓ Compare premature mortality (YPLL)
- ✓ Analyse mortality trends
- ✓ Support healthcare planning



Source

Department of Veterans Affairs
Office of Mental Health and
Suicide Prevention



Dataset

National Veteran All-Cause
Mortality Dataset
2018–2023



Included

2 Veteran populations
51 causes of death
6 years
612 observations



Variables

Deaths
Crude Rate
Age-adjusted Rate
YPLL
Percentages

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Methodology

Analytical Approach



Because the dataset contains aggregated population-level mortality statistics, predictive modelling was not appropriate.

The final analysis therefore used:

- ✓ Descriptive Statistics
- ✓ Comparative Analysis
- ✓ Longitudinal Trend Analysis
- ✓ Years of Potential Life Lost (YPLL)
- ✓ Data Visualisation

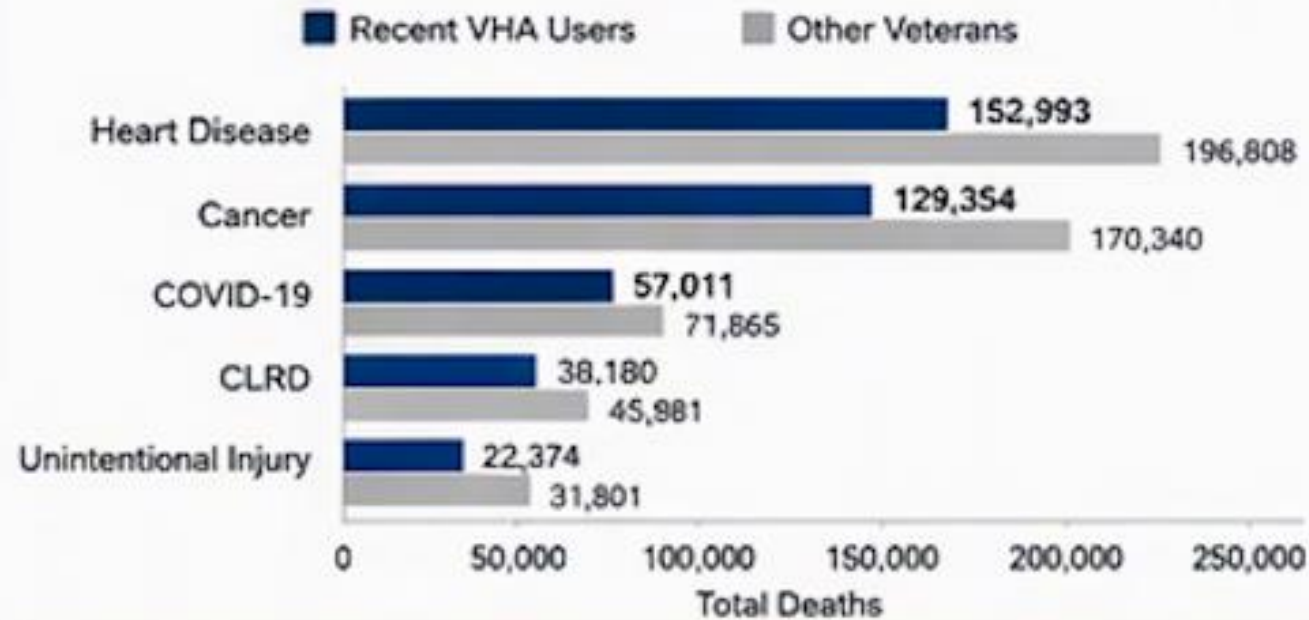


Regression and predictive modelling were not used because the dataset does not contain independent patient-level observations or predictor variables.

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Analysis 1 Comparative Mortality Burden

Leading Causes of Death (2018–2023)
Total Deaths

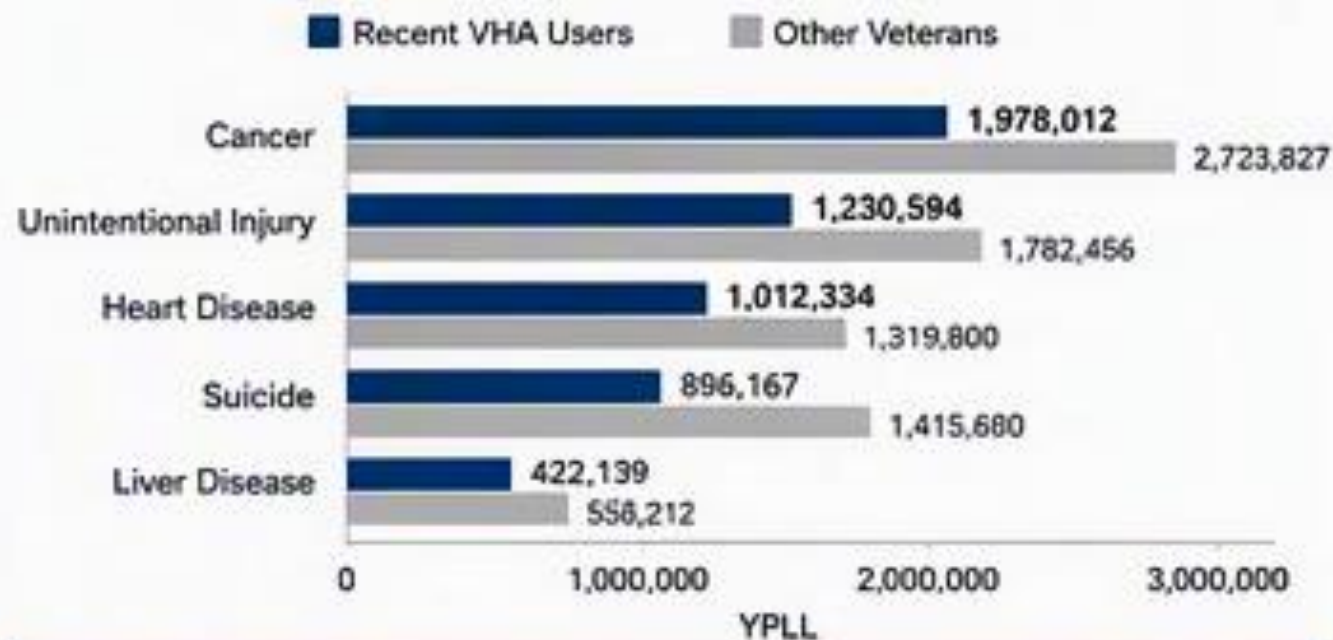


- 1 Heart disease remained the leading cause of death.
- 2 Cancer ranked second.
- 3 Other Veterans experienced higher cumulative mortality.

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Analysis 2 Premature Mortality

Years of Potential Life Lost (YPLL) (2018–2023)
Total YPLL

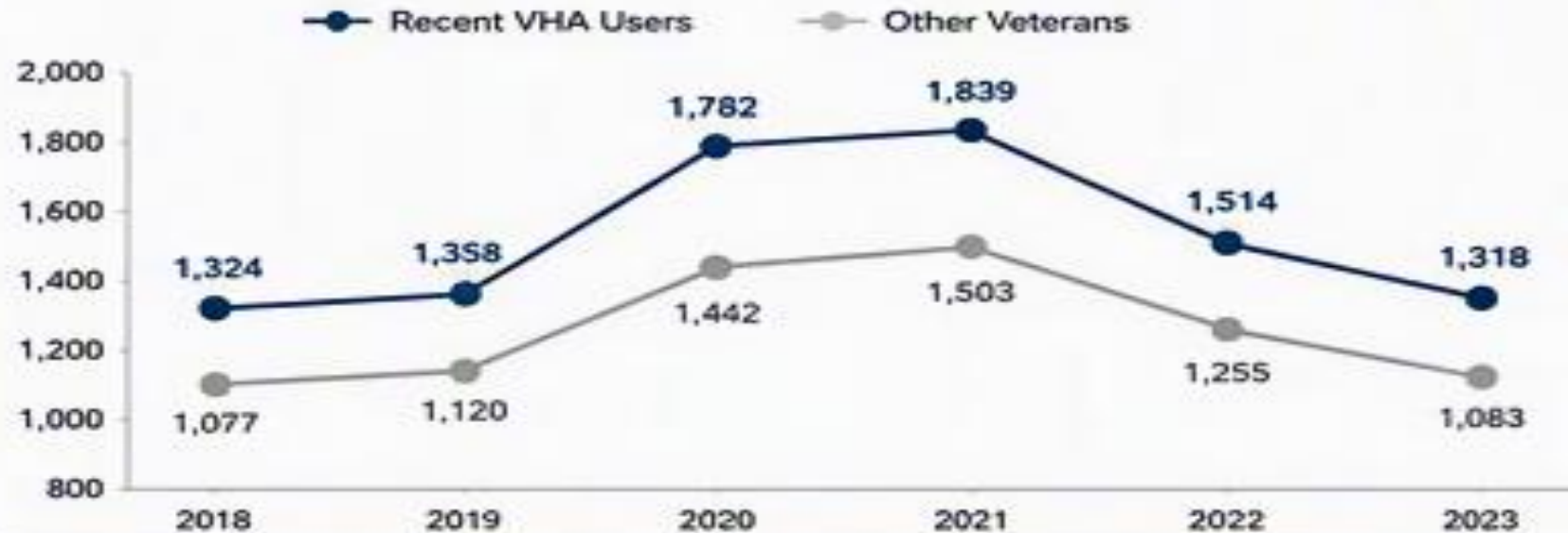


- 1 Cancer produced the greatest YPLL.
- 2 Suicide ranked fourth.
- 3 YPLL revealed a different perspective from mortality counts.

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Analysis 3 Mortality Trends

Age-Adjusted Mortality Rate (per 100,000)



- 1 Stable before COVID
- 2 Peak in 2020–2021
- 3 Decline after pandemic

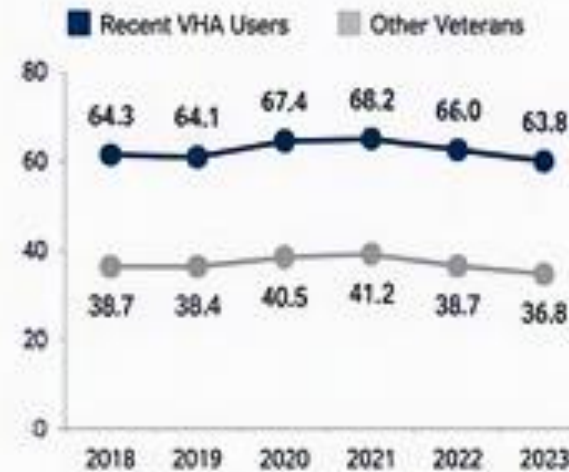
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Analysis 4 Suicide Analysis

Suicide Deaths



Suicide Age-Adjusted Mortality Rate (per 100,000)



- 1 Deaths remained relatively stable.
- 2 YPLL remained consistently high.
- 3 Recent VHA users had higher age-adjusted suicide mortality rates.

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Key Findings



Chronic diseases remain the largest mortality burden.



Suicide contributes disproportionately to premature mortality.



Both veteran populations experienced similar mortality trends.



YPLL complements mortality counts.

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Recommendations

Healthcare organisations should:



Continue investing in suicide prevention.



Monitor YPLL alongside mortality counts.



Strengthen chronic disease prevention.



Continue population health surveillance.



Use data analytics to support evidence-based decisions.

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Limitations



Public dataset



Aggregated data



No patient-level records



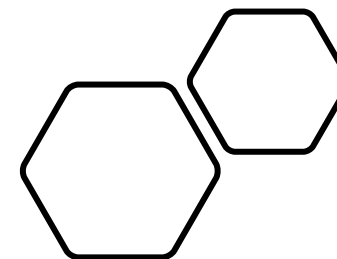
No regression analysis



No causal inference



Limited demographics



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Conclusion



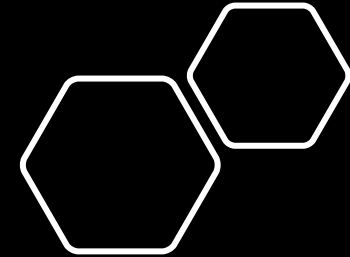
Although chronic diseases remain the leading causes of veteran mortality, analysing Years of Potential Life Lost alongside mortality counts demonstrates that suicide continues to impose a substantial burden of premature mortality. Descriptive population health analytics provide meaningful evidence to support healthcare planning and suicide prevention within the Veterans Health Administration.



Thank You



Questions?



- **References**

- Centers for Disease Control and Prevention. (2025). *CDC WONDER*. <https://wonder.cdc.gov/>
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- U.S. Department of Veterans Affairs. (2024). *National Veteran suicide prevention annual report: 2024 data appendix—All-cause mortality (2018–2023)*. https://www.mentalhealth.va.gov/suicide_prevention/data.asp
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- World Health Organization. (2019). *International statistical classification of diseases and related health problems (11th ed.)*. <https://icd.who.int/>

- **Source:** Author's analysis of the U.S. Department of Veterans Affairs *National Veteran Suicide Prevention Annual Report: 2024 Data Appendix - All - Cause Mortality (2018-2023)*.